

CHAPTER 17: LOCOMOTION AND MOVEMENT

Complete Textbook Coverage • NCERT Pages 217–229 • Landscape Study Guide

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Chapter Mind Map: Locomotion and Movement Hierarchy

1. Types of Movements

- **Amoeboid:** Macrophages, Leucocytes (Pseudopodia, Microfilaments)
- **Ciliary:** Tracheal lining (dust removal), Oviducts (Ova passage)
- **Muscular:** Limbs, Jaws, Tongue (Contractile properties)
- **Flagellar:** Sperm, Sponges canal, *Euglena*

2. Muscle Types & Mechanism

- **Skeletal:** Striated, Voluntary, Fascicles + Fascia
- **Visceral:** Smooth, Involuntary, Visceral organs
- **Cardiac:** Striated, Branched, Involuntary
- **Sarcomere:** Z-line, A-band, I-band, H-zone
- **Mechanism:** Sliding Filament Theory (Ca^{2+} + Troponin)

3. Skeletal System (206 Bones)

- **Axial (80):** Skull (22), Hyoid (1), Ear Ossicles (6), Vertebrae (26), Sternum (1), Ribs (24)
- **Appendicular (126):** Limbs ($30 \times 4 = 120$), Pectoral Girdle (4), Pelvic Girdle (2)

4. Joints & Clinical Disorders

- **Fibrous:** Immovable (Skull sutures)
- **Cartilaginous:** Slightly movable (Vertebrae)
- **Synovial:** Freely movable (Ball & Socket, Hinge, Pivot, Gliding, Saddle)
- **Disorders:** Myasthenia Gravis, Dystrophy, Tetany, Arthritis, Osteoporosis, Gout

Movement vs. Locomotion

Movement is a primary characteristic feature of all living organisms, ranging from unicellular streaming of protoplasm in *Amoeba* to movement of cilia, flagella, tentacles, and limbs. When voluntary movements result in a change of place or location, it is termed **Locomotion** (e.g., walking, running, climbing, flying, swimming).

🔑 Core Biological Golden Rule:

"All locomotions are movements, but all movements are not locomotions."

Ecological Significance of Locomotion:

Animals locomote in search of food, shelter, mate, suitable breeding grounds, favourable climatic conditions, or to escape from predators/enemies.

Three Main Types of Body Cell Movements

1. Amoeboid Movement:

Exhibited by specialized cells like **macrophages** and **leucocytes** in human blood. Effected by pseudopodia formed by protoplasmic streaming. Cytoskeletal elements like **microfilaments** are actively involved.

2. Ciliary Movement:

Occurs in internal tubular organs lined by ciliated epithelium. Coordinated ciliary sweeps in the **trachea** clear dust and foreign particles. Also facilitates passage of ova through the **female reproductive tract (fallopian tubes)**.

3. Muscular Movement:

Required for movements of limbs, jaws, tongue, etc. Relies on the contractile properties of muscles. Requires perfect coordination between **muscular, skeletal, and neural systems**.

Origin

Mesodermal origin

Body Mass Share

40% – 50% of adult body weight

Core Functional Properties

Excitability, Contractility, Extensibility, Elasticity

Flagellar Outgrowths

Spermatozoa swimming, Sponges canal, *Euglena*

Comprehensive Comparison Table: Three Types of Muscle Tissues

Property / Feature	Skeletal Muscle	Visceral (Smooth) Muscle	Cardiac Muscle
Location	Attached closely to skeletal bones	Inner walls of hollow visceral organs (digestive & genital tracts)	Wall of Heart (Myocardium)
Microscopic Appearance	Striated (Alternate dark A & light I bands)	Non-striated (Smooth appearance, fusiform)	Striated with intercalated discs
Nervous Control	Voluntary (Under conscious somatic nervous control)	Involuntary (Autonomic nervous control)	Involuntary (Autonomic nervous system influence)
Cellular Structure	Multinucleated Syncytium with Sarcolemma	Uninucleate, spindle-shaped cells	Branched cellular network with intercalated discs
Primary Functions	Locomotion, body posture changes	Peristalsis, food & gamete transport	Rhythmic contraction pumping blood throughout life

Detailed Anatomy of a Skeletal Muscle Bundle

- **Fascicles:** Each skeletal muscle is composed of multiple muscle bundles called *fascicles*.
- **Fascia:** Fascicles are held together by a common collagenous connective tissue layer named *fascia*.
- **Muscle Fibre (Cell):** Each fascicle contains numerous muscle fibres. The cell membrane is the **Sarcolemma**, enclosing **Sarcoplasm**.
- **Syncytium:** Muscle fibre is a syncytium because sarcoplasm contains multiple nuclei.
- **Sarcoplasmic Reticulum:** Specialized endoplasmic reticulum acting as the primary storehouse of Calcium ions (Ca^{2+}).
- **Myofibrils / Myofilaments:** Parallellly arranged filaments in sarcoplasm exhibiting alternate dark (**A-band**) and light (**I-band**).

Figure 17.1 — Cross-Sectional View of Muscle & Fascicles

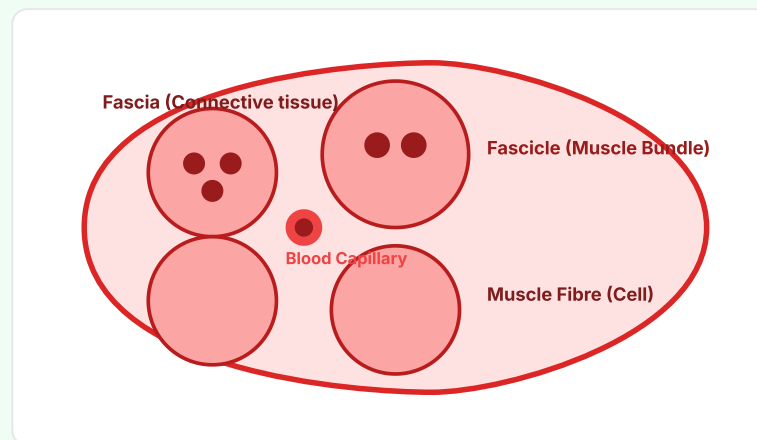
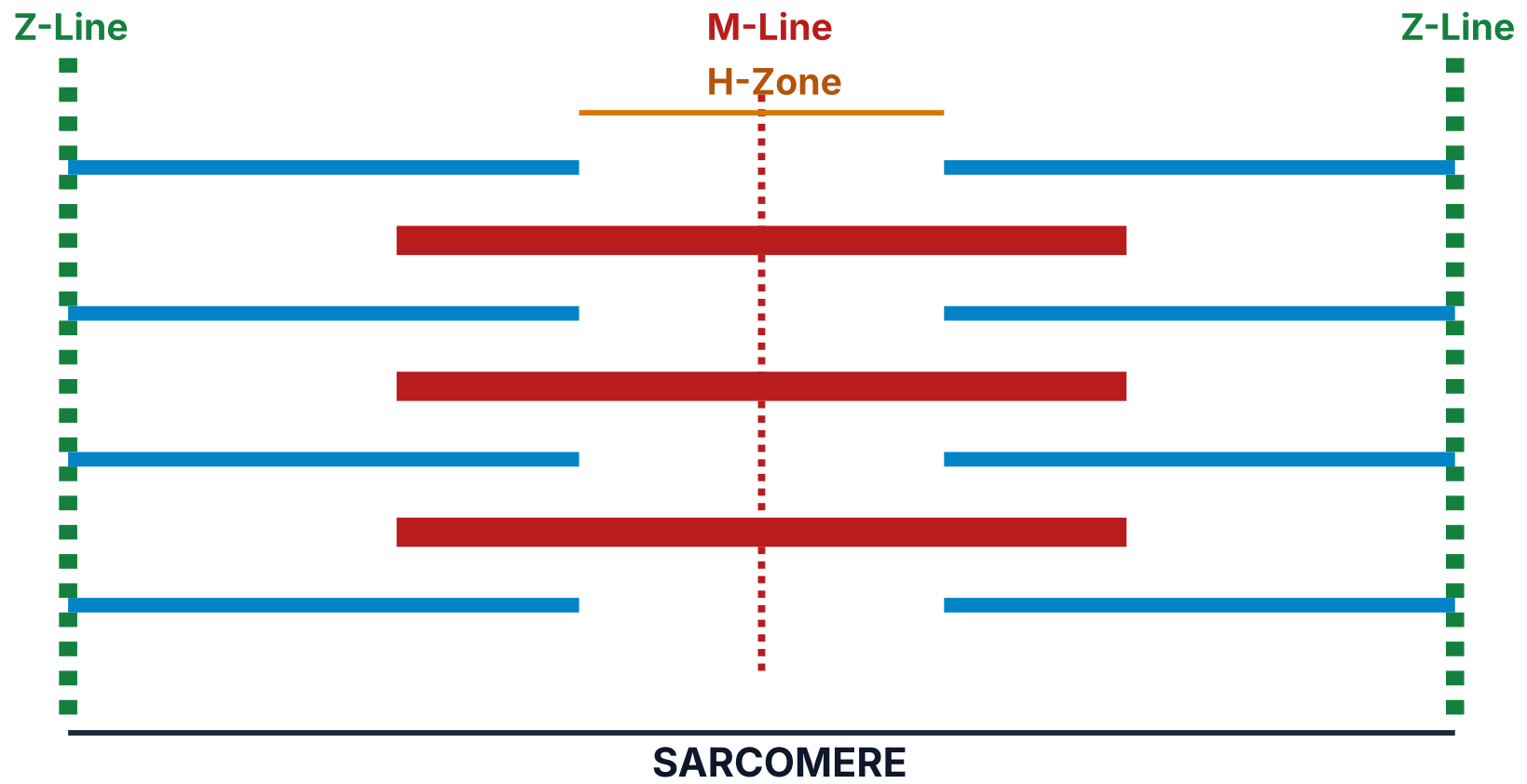


Figure 17.2 — Detailed Anatomy of Sarcomere



Sarcomere Functional Regions & Formulas

- **I-Band (Isotropic Band):** Light band containing only **Actin (thin) filaments**. Bisected by elastic fibre called the **Z-line**.
- **A-Band (Anisotropic Band):** Dark band containing **Myosin (thick) filaments** along with overlapping thin filaments. Held in middle by thin fibrous **M-line**.
- **H-Zone:** Central part of thick filament *not overlapped* by thin filaments in resting state.
- **Sarcomere Definition:** The portion of myofibril between two successive Z-lines; considered the functional unit of contraction.

Sarcomere Structural Mathematical Formula:

$$1 \text{ Sarcomere} = \frac{1}{2}(\text{I-Band}) + 1 \text{ Full A-Band} + \frac{1}{2}(\text{I-Band})$$

$$\text{Resting H-Zone} = \text{Length of A-Band} - \text{Overlapping Thin Filaments}$$

17.2.1 Structure of Contractile Proteins (Actin & Myosin)

● Thin Filament (Actin) Components

- **F-Actin (Filamentous):** Two 'F' actins helically wound together. Each F-actin is a polymer of monomeric '**G**' (**Globular**) actins.
- **Tropomyosin:** Two filaments of tropomyosin run close to F-actins throughout its length.
- **Troponin:** Complex protein distributed at regular intervals on tropomyosin. In resting state, a subunit of troponin **masks active binding sites for myosin** on actin filaments.

● Thick Filament (Myosin) Components

- **Meromyosins:** Myosin is a polymerised protein made of monomeric units called Meromyosins.
- **Heavy Meromyosin (HMM):** Consists of Globular Head + Short Arm (forms **Cross Arm** projecting outwards).
- **Light Meromyosin (LMM):** Constitutes the Tail portion.
- **Active ATPase Head:** Globular head has active **ATPase enzyme sites**, ATP binding sites, and active sites for actin.



17.2.2 MECHANISM OF MUSCLE CONTRACTION (SLIDING FILAMENT THEORY)

NCERT PAGES 221–223

Sequential Biochemical & Biophysical Steps in Muscle Contraction

1

CNS Neural Signal

Signal sent via motor neuron to **Neuromuscular Junction (Motor-end plate)**. Releases **Acetylcholine (ACh)**.

2

Action Potential & Ca^{2+}

Action potential spreads through sarcolemma. Causes release of Ca^{2+} **ions** from sarcoplasmic reticulum into sarcoplasm.

3

Unmasking Active Sites

Ca^{2+} binds subunit of **Troponin**, removing masking from active myosin binding sites on actin filaments.

4

Cross Bridge Power Stroke

Myosin head hydrolyses ATP ($\text{ADP} + \text{P}_i$), binds actin to form **Cross Bridge**, pulling Z-lines inward toward center of A-band.

5

Relaxation Phase

New ATP binds, cross-bridge breaks. Ca^{2+} pumped back to sarcoplasmic cisternae. Troponin re-masks actin sites.



Key Biochemical Equations of Contraction & Fatigue:

1. ATP Hydrolysis by Myosin Head:



2. Anaerobic Glycogen Breakdown (Muscle Fatigue):



Structural Band Alterations During Contraction

✓ **I-Band:** Gets *reduced/shortened* as thin filaments slide inwards.

✓ **H-Zone:** Reduces and completely *disappears* upon maximal contraction.

✓ **Sarcomere:** Overall length *shortens* (Z-lines pulled together).

⚠ **A-Band: RETAINS ITS ORIGINAL LENGTH** (Does not shorten!).

Comparison: Red Muscle Fibres vs. White Muscle Fibres

Feature	Red Fibres (Aerobic)	White Fibres (Anaerobic)
Myoglobin Content	Very High (Reddish appearance)	Very Low (Pale or whitish appearance)
Mitochondria	Plenty (Utilises O ₂ for ATP)	Few in number
Sarcoplasmic Reticulum	Moderate amount	Very High amount
Metabolism	Aerobic process (Fatigue resistant)	Anaerobic process (Fatigues rapidly)

17.3 SKELETAL SYSTEM (206 BONES)

NCERT PAGES 224–226

Bone (Hard Connective Tissue)

Has a very hard, non-pliable matrix due to high concentration of **Calcium salts** (Calcium phosphate & carbonate).

Cartilage (Pliable Connective Tissue)

Has a slightly pliable matrix due to **Chondroitin salts**.

 Total Human Skeleton Calculation Formula:

$$\text{Total Human Bones (206)} = \text{Axial Skeleton (80)} + \text{Appendicular Skeleton (126)}$$

A. Axial Skeleton Breakdown (80 Bones)

Distributed along central body axis

1. Skull (22 Bones)

- **Cranial (8):** Frontal, Parietal (2), Temporal (2), Occipital, Sphenoid, Ethmoid.
- **Facial (14):** Form front skull part (Nasal, Maxilla, Zygomatic, Mandible, etc.).
- **Dicondylic Skull:** Articulates with atlas vertebra via 2 Occipital Condyles.

2. Hyoid & Ear Ossicles (7 Bones)

- **Hyoid Bone (1):** Single U-shaped bone present at base of buccal cavity.
- **Ear Ossicles (6 = 3 × 2):** Malleus, Incus, Stapes in middle ears.

3. Vertebral Column (26 Units)

- **Cervical (7):** Neck region ($C_1 - C_7$). $C_1 = \text{Atlas}$.
- **Thoracic (12):** $T_1 - T_{12}$.
- **Lumbar (5):** $L_1 - L_5$.
- **Sacral (1 fused):** 5 fused into 1.
- **Coccygeal (1 fused):** 4 fused into 1.

4. Sternum & Ribs (25 Bones)

- **Sternum (1):** Flat bone on ventral midline of thorax.
- **True Ribs (Pairs 1–7):** Dorsally attached to thoracic vertebrae, ventrally to sternum via hyaline cartilage.
- **False Ribs (Pairs 8–10):** Vertebrochondral ribs; join 7th rib via hyaline cartilage.
- **Floating Ribs (Pairs 11–12):** Not connected ventrally.



Memory Trick for Cranial Bones (8 Bones): Use mnemonic "**PEST OF**"
Parietal (2), Ethmoid (1), Sphenoid (1), Temporal (2), Occipital (1), Frontal (1) = 8 Bones total.

B. Appendicular Skeleton Breakdown (126 Bones)

Limbs & Girdles

Forelimbs (60 Bones) & Pectoral Girdle (4 Bones)

- **Humerus:** 1 per arm (Upper arm bone)
 - **Radius & Ulna:** 2 per arm (Forearm)
 - **Carpals (Wrist):** 8 per arm
 - **Metacarpals (Palm):** 5 per arm
 - **Phalanges (Digits):** 14 per arm (Formula: 2, 3, 3, 3, 3)
-
- **Pectoral Girdle (2 halves = 4 bones):** Each half consists of 1 **Clavicle** (collar bone with 2 curvatures) and 1 **Scapula** (flat triangular bone with spine, acromion process, and **Glenoid Cavity** articulating with humerus head).

Hindlimbs (60 Bones) & Pelvic Girdle (2 Bones)

- **Femur (Thigh):** 1 per leg (Longest bone of body)
 - **Patella (Knee cap):** 1 per leg (Cup-shaped ventral knee cover)
 - **Tibia & Fibula:** 2 per leg (Shank bones)
 - **Tarsals (Ankle):** 7 per leg
 - **Metatarsals:** 5 per leg
 - **Phalanges (Digits):** 14 per leg
-
- **Pelvic Girdle (2 Coxal bones):** Formed by fusion of **Ilium, Ischium, and Pubis**. Features deep **Acetabulum cavity** for femur head. Joined ventrally by **Pubic Symphysis** (fibrous cartilage).

17.4 Classification of Joints

Joints are points of contact between bones, or between bones and cartilages. Force generated by muscles is used to carry out movement through joints, where the joint acts as a **fulcrum**.

Joint Category	Structural Feature	Degree of Movement	Textbook Examples
1. Fibrous Joints	Bones fused end-to-end with dense fibrous connective tissue	No movement (Immovable)	Sutures of flat skull bones forming cranium
2. Cartilaginous Joints	Bones joined together with the help of cartilages	Permits limited movements	Joints between adjacent vertebrae in vertebral column
3. Synovial Joints	Fluid-filled synovial cavity between articulating surfaces	Considerable movement (Freely movable)	• Ball & Socket: Humerus & Pectoral girdle • Hinge Joint: Knee & Elbow joint • Pivot Joint: Between Atlas (C₁) and Axis (C₂) • Gliding Joint: Between Carpals • Saddle Joint: Between Carpal & Metacarpal of thumb

17.5 Clinical Disorders of Muscular and Skeletal Systems

1. Myasthenia Gravis

Autoimmune disorder affecting neuromuscular junction. Leads to fatigue, weakening, and paralysis of skeletal muscle.

2. Muscular Dystrophy

Progressive degeneration of skeletal muscle, mostly due to a **genetic disorder**.

3. Tetany

Rapid spasms (wild contractions) in muscle due to **low Ca²⁺ levels** in body fluid.

4. Arthritis

Inflammation of joints causing severe pain and stiffness.

5. Osteoporosis

Age-related disorder characterized by decreased bone mass and increased risk of fractures. Common cause: **Decreased levels of estrogen**.

6. Gout

Inflammation of joints due to accumulation of **uric acid crystals**.



Complete NCERT Textbook Exercise Solutions

Q1. Draw the diagram of a sarcomere of skeletal muscle showing different regions.

Answer: Refer to *Figure 17.2* above in notes. Key labelled regions include: **Z-line** (bisects I-band), **I-band** (Actin thin filaments only), **A-band** (Myosin thick filaments), **H-zone** (un-overlapped central myosin zone), and **M-line** (central fibrous line).

Q2. Define sliding filament theory of muscle contraction.

Answer: The sliding filament theory states that contraction of a muscle fibre takes place by the sliding of the thin (actin) filaments over the thick (myosin) filaments towards the center of the A-band, resulting in the shortening of the sarcomere.

Q3. Describe the important steps in muscle contraction.

Answer: 1. CNS sends motor nerve signal → Acetylcholine release at motor end plate. 2. Action potential in sarcolemma causes Ca^{2+} release from sarcoplasmic reticulum. 3. Ca^{2+} binds Troponin, unmasking myosin sites on actin. 4. Myosin head hydrolyses ATP, forms cross-bridge, and pulls actin filaments inward (power stroke). 5. New ATP binds, breaking cross-bridge; Ca^{2+} pumped back for relaxation.

Q4. Write True or False. If false, change the statement so that it is true.

- (a) Actin is present in thin filament. → **True**
- (b) H-zone of striated muscle fibre represents both thick and thin filaments. → **False**. *Corrected: H-zone represents only thick filaments not overlapped by thin filaments.*
- (c) Human skeleton has 206 bones. → **True**
- (d) There are 11 pairs of ribs in man. → **False**. *Corrected: There are 12 pairs of ribs in humans.*
- (e) Sternum is present on the ventral side of the body. → **True**

Q5. Write differences between: (a) Actin & Myosin (b) Red & White muscles (c) Pectoral & Pelvic girdle

Answer: Refer to detailed comparison tables structured in Sections 17.2 and 17.3 above.

Q6. Match Column I with Column II:

Answer: (a) Smooth muscle → **(iv) Involuntary** | (b) Tropomyosin → **(ii) Thin filament** | (c) Red muscle → **(i) Myoglobin** | (d) Skull → **(iii) Sutures**

Q9. Name the type of joint between the following:

(a) Atlas / Axis → **Pivot Joint**

(b) Carpal / Metacarpal of thumb → **Saddle Joint**

(c) Between Phalanges → **Hinge Joint**

(e) Between Cranial Bones → **Fibrous Joint (Sutures)**

(d) Femur / Acetabulum → **Ball & Socket Joint**

(f) Between Pubic Bones in Pelvic Girdle → **Cartilaginous Joint (Pubic Symphysis)**

Q10. Fill in the blank spaces:

(a) All mammals have **7** cervical vertebrae. (b) Number of phalanges in each limb is **14**. (c) Thin filament contains 2 'F' actins and **Troponin** and **Tropomyosin**. (d) In muscle fibre Ca^{2+} is stored in **Sarcoplasmic Reticulum**. (e) **11th** and **12th** pairs of ribs are floating ribs. (f) Human cranium is made of **8** bones.



FINAL REVISION & EXAM CHECKLIST

Key Formulas & Counts

- Total Bones = 206 (Axial 80 + Appendicular 126)
- Vertebral Formula = $\text{C}_7\text{T}_{12}\text{L}_5\text{S}_{(5)}\text{Co}_{(4)}$
- Sarcomere = Z-line to Z-line

Band Changes in Contraction

- I-Band: Shortens
- H-Zone: Disappears
- A-Band: NO CHANGE IN LENGTH!

High-Yield Clinical Pearls

- Low Ca^{2+} in fluid → Tetany
- Low Estrogen → Osteoporosis
- Uric Acid Crystals → Gout
- Autoimmune Neuromuscular → Myasthenia Gravis